

Maternal Health Risk Stratification

Statistical Analysis, Exploratory Data Analysis, and Interpretable Machine Learning

StudyBuild | Health and Medical Data Science Project

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Abstract

This report examines the UCI Maternal Health Risk dataset using descriptive statistics and an interpretable classification model. The project CSV contains 1,014 observations, whereas the UCI description reports 1,013 instances. The analysis therefore uses the supplied file and records this discrepancy explicitly. No missing values were found. The raw data contained 562 exact repeated rows and two observations with HeartRate = 7 bpm. For the primary modeling analysis, the two implausible heart rate records and exact repeated rows were excluded, leaving 451 unique observations. Because patient identifiers are not available, removal of repeated rows is treated as a methodological assumption rather than evidence that the rows belong to the same patient. Exploratory analysis showed the clearest separation for Blood Sugar, followed by systolic and diastolic blood pressure. The absolute standardized mean differences were 1.61 for Blood Sugar, 0.70 for SystolicBP, and 0.61 for DiastolicBP. A multinomial Logistic Regression model was fitted after an 80/20 stratified split, with StandardScaler applied within the modeling pipeline. Test accuracy was 0.670. Recall was 0.565 for High Risk, 1.000 for Low Risk, and 0.048 for Mid Risk. All ten High Risk observations missed by the model were classified as Low Risk. These results indicate that overall accuracy alone is not sufficient for evaluating performance in this dataset. The model should be viewed as an educational baseline and not as a diagnostic or clinical decision tool.

Keywords: maternal health; risk stratification; blood sugar; blood pressure; logistic regression; recall; error analysis; interpretable machine learning.

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1. Introduction

Maternal health risk stratification requires careful analysis because physiological measurements vary across individuals and because errors do not have equal practical importance. Missing a true High Risk observation may be more consequential than confusing two lower risk categories. For this reason, the project focuses on transparent exploratory analysis, a simple interpretable classifier, class specific evaluation, and careful interpretation rather than accuracy alone.

The main analytical question is whether the recorded physiological measurements differ meaningfully across Low, Mid, and High Risk groups, and how well a simple multinomial Logistic Regression model can distinguish these three categories.

The analysis begins with data quality assessment, followed by exploratory summaries and standardized group comparisons. The data are then divided into stratified training and test sets, the model is fitted, classification errors are examined, coefficients are interpreted, and the results are discussed in light of the study limitations.

2. Data and Study Design

2.1 Data source and variables

The project uses the UCI Maternal Health Risk dataset, Dataset 863, which was collected in maternal health settings in rural Bangladesh through an Internet of Things monitoring system. The StudyBuild brief reports 1,013 instances in the official UCI version, while the CSV supplied for this analysis contains 1,014 rows. The report uses the supplied project file and records this discrepancy as a source version issue.

Variable	Unit / Type	Description
Age	years	Maternal age
SystolicBP	mmHg	Systolic blood pressure
DiastolicBP	mmHg	Diastolic blood pressure
BS	mmol/L	Blood sugar
BodyTemp	°F	Body temperature
HeartRate	bpm	Heart rate
RiskLevel	categorical	Low, Mid, or High Risk

2.2 Data quality decisions and analytical sample

The raw file contained 1,014 observations and seven variables, with no missing values. The main findings from the data quality review and the decisions used in the analysis are summarized below.

Check	Finding	Decision
Missing values	0	No imputation required.
Exact repeated rows	562	Removed for the primary modeling analysis to reduce the chance that identical rows appear in both train and test sets. Because patient IDs are unavailable, this is an analytical assumption.
HeartRate = 7 bpm	2 rows	Excluded as clinically implausible measurements.
Very young ages	Minimum age = 10	Retained because there was insufficient evidence to classify them as data entry errors.
Conflicting labels	35 identical feature profiles	Retained. Same predictors with different RiskLevel labels may indicate label ambiguity or unmeasured clinical information.
Final modeling sample	451 observations	Used for EDA, SMD analysis, training and test split, and Logistic Regression.

Removing exact repeated rows changed the class distribution noticeably. Low Risk increased from 40.0% of the raw data to 51.7% of the modeling sample, while Mid Risk decreased from 33.1% to 23.5%. Repeated rows were therefore not distributed evenly across risk categories. This change is important and is reported as part of the preprocessing assumptions.

3. Exploratory Data Analysis

3.1 Risk level distribution

Risk Level	Count	Percent
Low Risk	233	51.7%
Mid Risk	106	23.5%
High Risk	112	24.8%

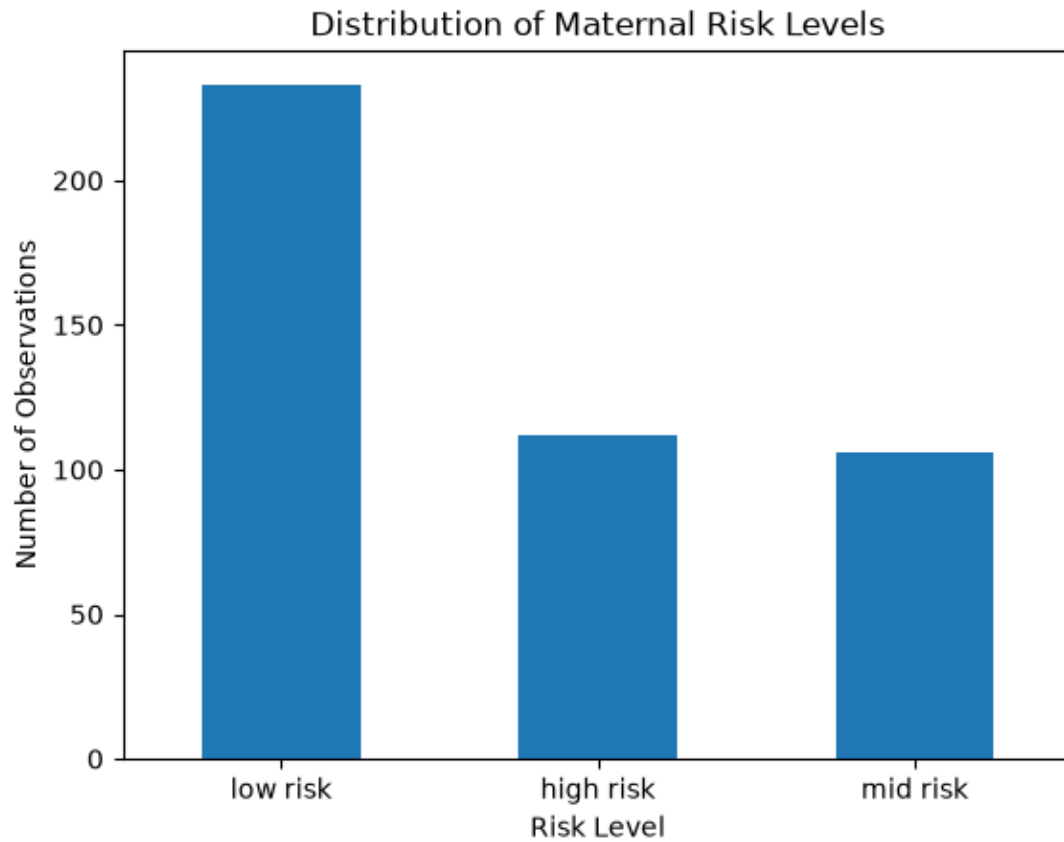


Figure 1. Distribution of Low, Mid, and High Risk observations in the modeling sample.

The three classes are moderately imbalanced, but none is extremely rare. Class specific metrics are still necessary because prevalence and classification difficulty differ across the three groups.

3.2 Clinical patterns across risk groups

Risk	Age	SBP	DBP	BS	BodyTemp	HeartRate
Low Risk	27.37	105.37	72.72	7.20	98.36	73.04
Mid Risk	28.54	112.41	74.89	7.89	98.86	73.90
High Risk	33.73	119.49	81.54	11.17	99.23	76.48

Mean values were higher in the High Risk group for all six recorded measurements. The largest descriptive difference was observed for Blood Sugar, followed by the blood pressure variables. Mean age also increased across the risk groups, although substantial overlap remained between individual observations.

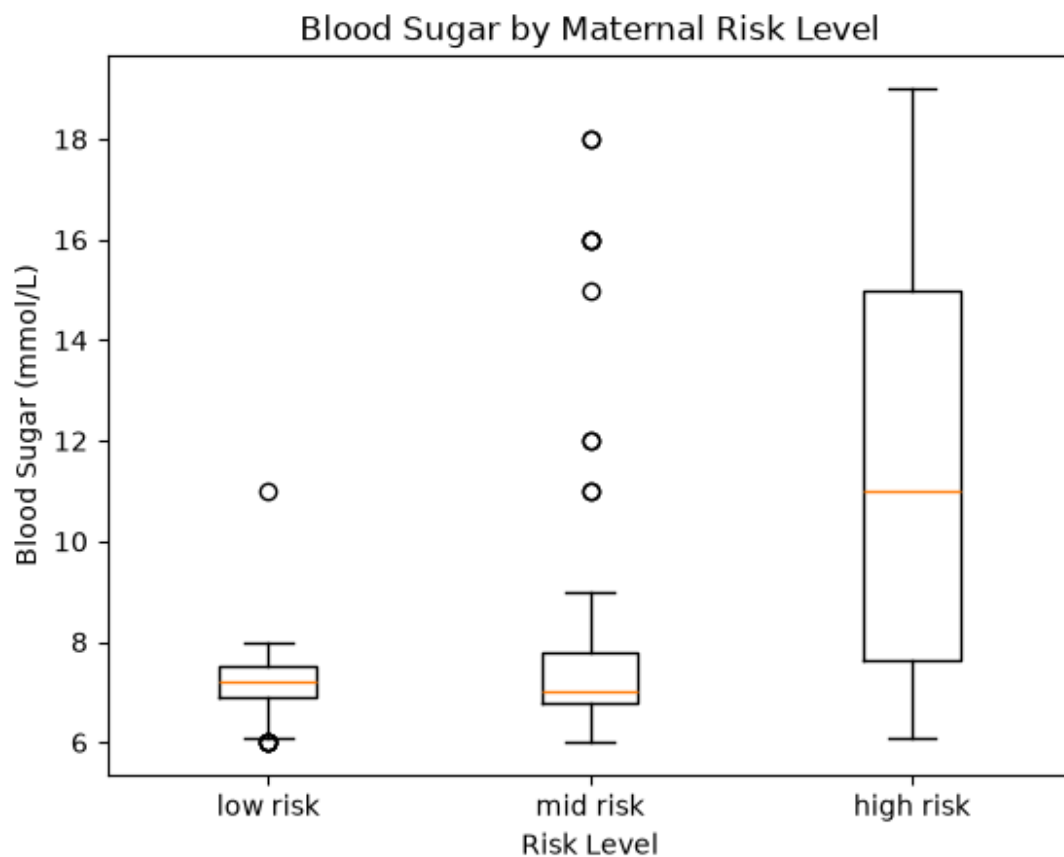


Figure 2. Blood Sugar distribution by maternal risk group.

Blood Sugar shows the clearest visual separation among the three groups. Values in the High Risk group are generally higher and more variable than those in the Low Risk group.

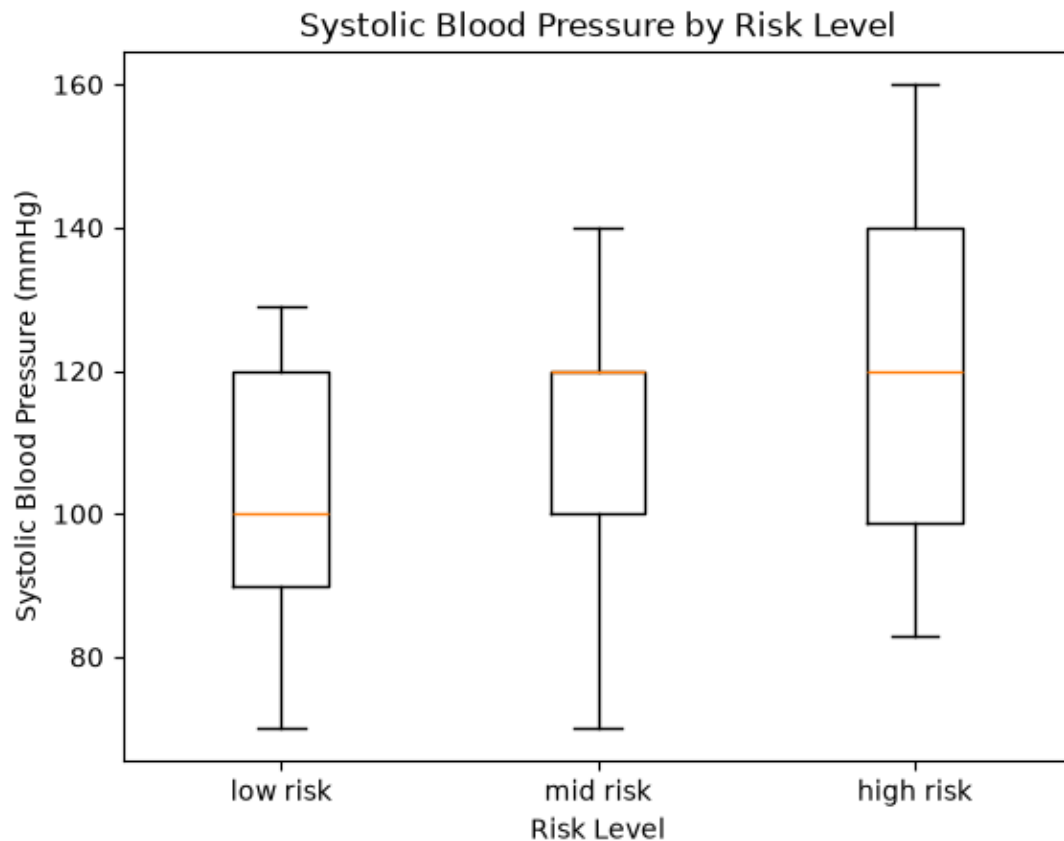


Figure 3. Systolic blood pressure distribution by maternal risk group.

Systolic blood pressure also increases across risk categories, but the distributions overlap considerably. This suggests that no single blood pressure threshold is sufficient to reproduce the three RiskLevel categories.

3.3 Association with High Risk using standardized mean differences

To compare variables measured on different scales, the High Risk group was contrasted with the combined Low and Mid Risk groups using absolute standardized mean differences, or SMDs. Each SMD expresses a mean difference relative to the pooled standard deviation, which makes the magnitude of separation comparable across age, blood pressure, blood sugar, temperature, and heart rate.

Clinical Feature	SMD : High Risk vs Other
Blood Sugar	1.613
Systolic BP	0.695
Diastolic BP	0.611
Body Temperature	0.515
Age	0.443
Heart Rate	0.428

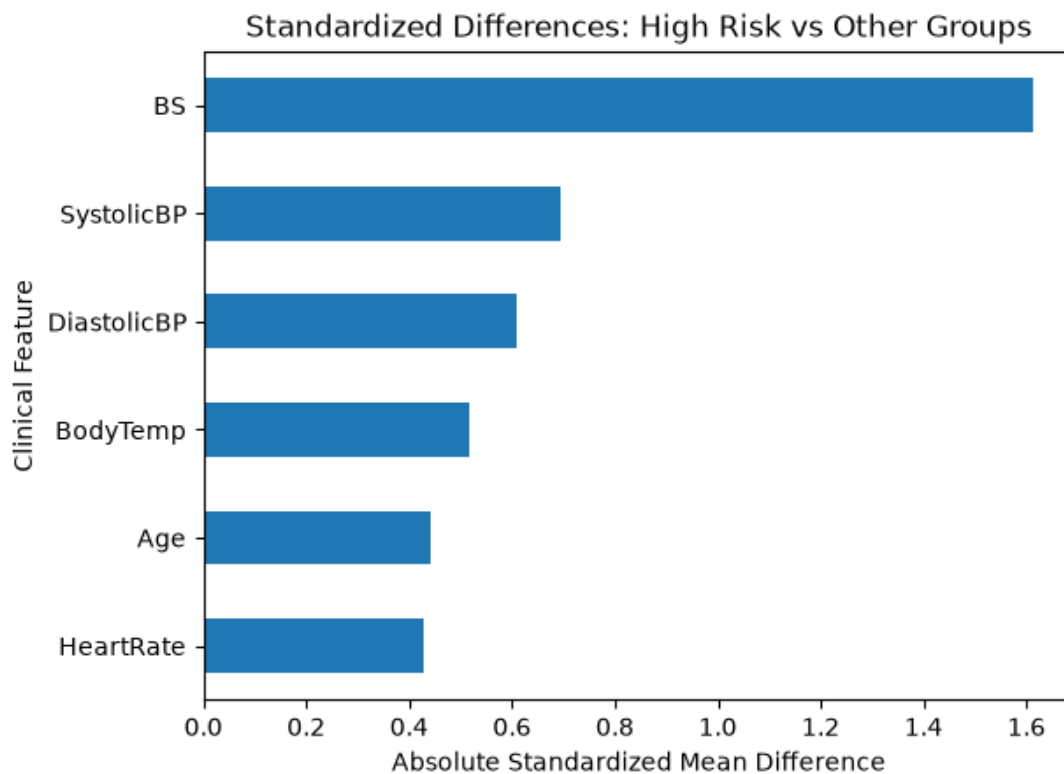


Figure 4. Absolute standardized mean differences for High Risk compared with Low and Mid Risk.

Blood Sugar had the largest standardized separation, followed by systolic and diastolic blood pressure. These results describe associations within the dataset and should not be interpreted as causal effects.

4. Statistical and Machine Learning Methods

4.1 Training and test design

The 451 observation modeling dataset was divided into 80% training data and 20% test data using `random_state = 42` and `stratify = y`. This produced 360 training observations and 91 test observations. Stratification kept the class proportions similar in the two subsets, with approximately 51.7% Low Risk, 25% High Risk, and 23% Mid Risk.

4.2 Logistic Regression pipeline

A multinomial Logistic Regression model was selected as the primary baseline because it is simple to interpret and appropriate for a three category outcome. `StandardScaler` and `LogisticRegression` were combined in a `scikit-learn Pipeline` so that scaling parameters were estimated from the training data and then applied to the test data. This approach limits information leakage during preprocessing.

```
Pipeline([
  ("scaler", StandardScaler()),
  ("logreg", LogisticRegression(max_iter=1000))
])
```

4.3 Evaluation strategy

Model performance was assessed using accuracy, precision, recall, F1 score, and a confusion matrix. Recall for High Risk received particular attention because it represents the proportion of true High Risk observations identified by the model. Macro averages were also examined because a single overall accuracy value can conceal weak performance in one class.

5. Machine Learning Results

5.1 Overall and class specific performance

Class	Precision	Recall	F1	Support
High Risk	0.812	0.565	0.667	23
Low Risk	0.635	1.000	0.777	47
Mid Risk	1.000	0.048	0.091	21

Overall test accuracy was 0.670 and macro F1 was 0.511. Performance differed substantially across classes. The model identified all Low Risk observations, but correctly identified only 56.5% of High Risk observations and 4.8% of Mid Risk observations.

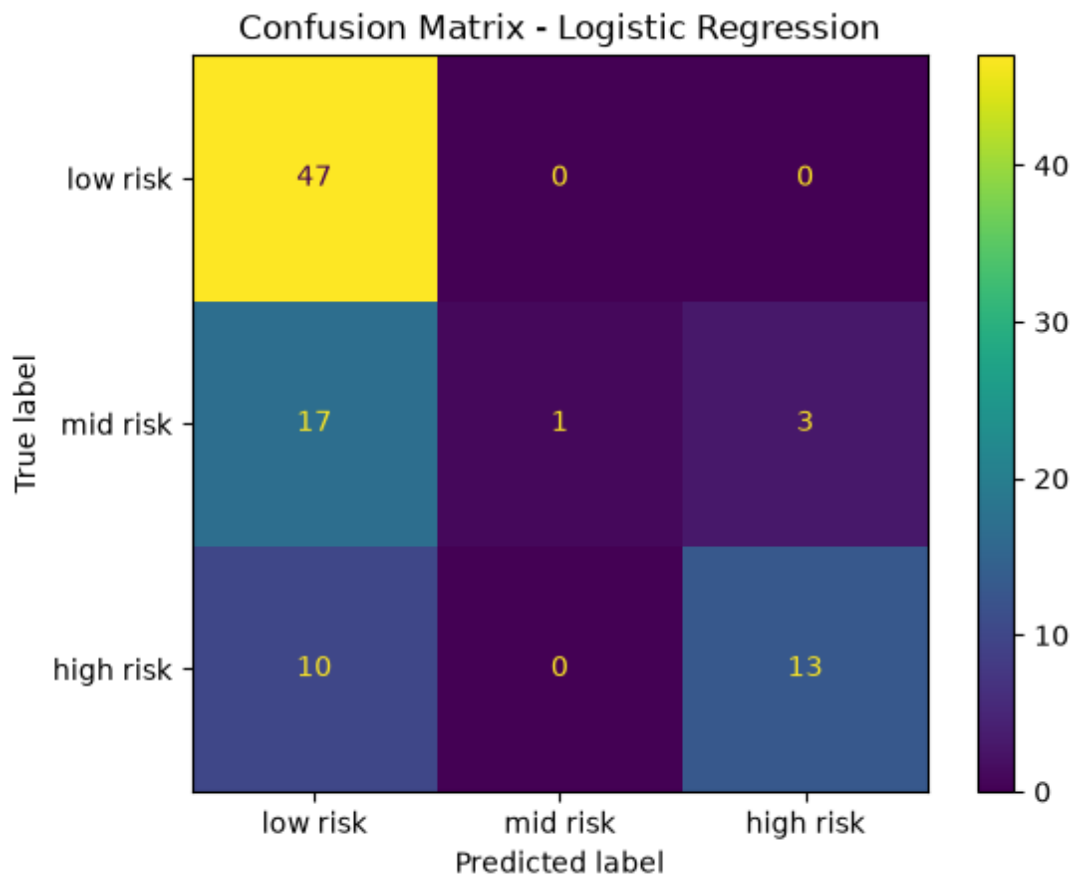


Figure 5. Confusion matrix for the multinomial Logistic Regression baseline.

Among the 23 true High Risk observations, 13 were correctly identified and 10 were missed. This corresponds to a false negative proportion of approximately 43.5% for High Risk in this test split. Mid Risk was even more difficult to identify, with only one of 21 observations classified correctly.

5.2 Error analysis

Error Type	Count
Mid Risk to Low Risk	17
High Risk to Low Risk	10
Mid Risk to High Risk	3

The model tended to assign uncertain observations to Low Risk. All ten missed High Risk observations were classified as Low Risk. Although this is an educational analysis rather than a clinical system, this error pattern is important because it shows where the baseline model performs poorly.

Feature	Missed High Risk	Correctly Identified High Risk
Age	33.30	36.77
SystolicBP	117.00	124.23
DiastolicBP	84.50	84.62
BS	7.35	12.81
BodyTemp	98.50	99.08
HeartRate	75.50	80.92

The clearest difference between missed and correctly identified High Risk observations was Blood Sugar. The mean value was 7.35 mmol/L among missed cases and 12.81 mmol/L among correctly identified cases. High Risk observations without markedly elevated Blood Sugar were therefore more difficult for the model to identify.

6. Feature Interpretation

Because the predictors were standardized before Logistic Regression was fitted, coefficient magnitudes can be compared more meaningfully across variables. These coefficients describe the fitted prediction model and should not be interpreted as causal clinical effects.

Feature	High Risk Standardized Coefficient
Blood Sugar	+1.176
Body Temperature	+0.524
Systolic BP	+0.345
Age	-0.214
Diastolic BP	+0.178
Heart Rate	+0.166

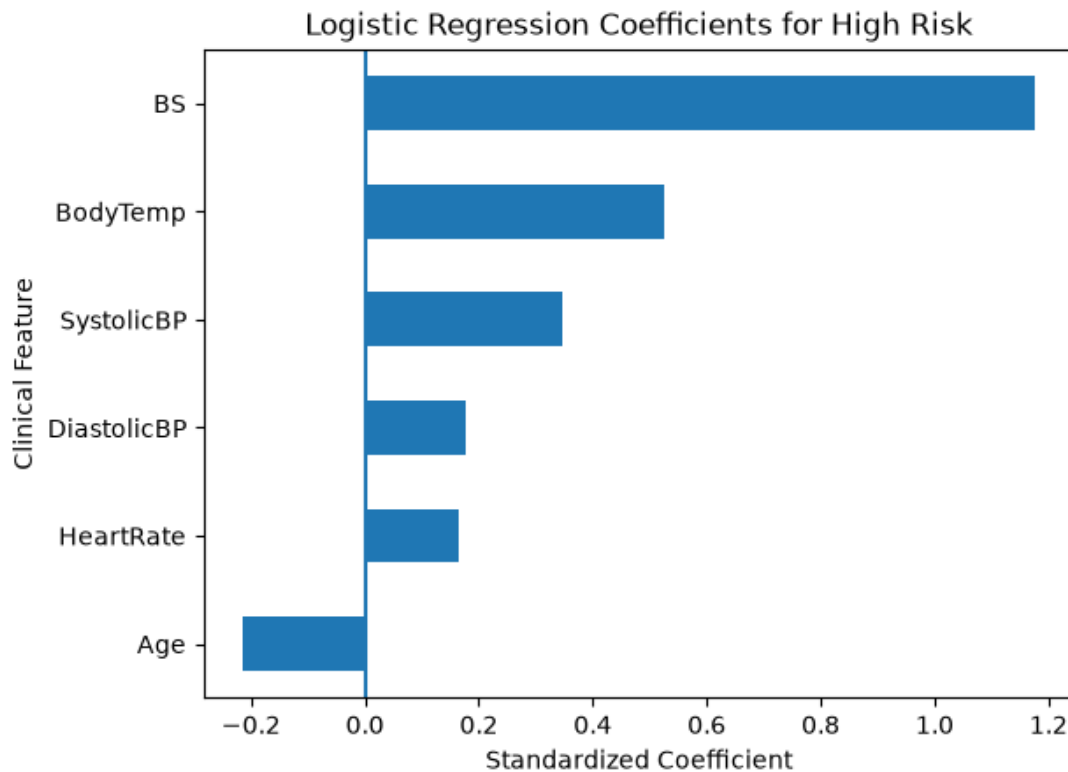


Figure 6. Standardized Logistic Regression coefficients for the High Risk class.

Blood Sugar had the largest positive coefficient for the High Risk class, followed by Body Temperature and Systolic BP. The negative coefficient for Age does not imply that older age is protective. In a multiclass Logistic Regression model, each coefficient is estimated while the other predictors remain in the model, so correlations among variables and overlap between classes can affect both magnitude and direction.

7. Discussion

The descriptive results and the fitted model point to the same general pattern. Blood Sugar provides the clearest separation between High Risk and the lower risk groups, while systolic and diastolic blood pressure add further information. Both the SMD analysis and the Logistic Regression coefficients identify Blood Sugar as the strongest signal among the available predictors.

The classification results also show the limits of a simple linear model for this three category problem. Mid Risk is poorly separated from Low Risk, and several High Risk observations with lower Blood Sugar values are missed. In addition, 35 identical predictor profiles are associated with more than one RiskLevel label. This suggests that the six recorded predictors do not fully explain how the risk category was assigned.

The error pattern provides more information than the 67% accuracy value alone. In health data analysis, it is important to examine which observations are missed, which measurements characterize those errors, and whether the available predictors contain enough information to distinguish the target categories.

8. Limitations

- The primary modeling sample contains 451 unique observations after the selected preprocessing steps.

- Only six clinical predictors are available; potentially important variables such as gestational age, obstetric history, comorbidities, laboratory results, medication use, and longitudinal measurements are absent.
- Patient identifiers are unavailable, so identical rows cannot be confirmed as repeated records from the same patient. Their removal is therefore a methodological assumption, and it changes the class proportions materially.
- Some identical predictor profiles have different RiskLevel labels. This may reflect label ambiguity, clinical information that is not included in the dataset, or limitations in the original labeling process.
- The project CSV contains 1,014 rows, while the UCI project description reports 1,013 instances. The available materials do not explain this difference between versions.
- The model was evaluated on a single stratified training and test split and has not been validated on an independent dataset.
- The target is an existing RiskLevel category rather than a prospectively observed clinical outcome.
- The observed associations and model coefficients do not establish causal clinical relationships.

9. Recommendations and Future Work

- Repeat the analysis as a sensitivity study using both the full dataset and the dataset with exact repeated rows removed.
- Use repeated stratified cross validation to assess how stable class specific performance is across different data splits.
- Collect or obtain richer clinical variables, particularly gestational age, obstetric history, comorbidities, laboratory measurements, and longitudinal records.
- Investigate label provenance to determine how RiskLevel was originally assigned and whether the predictors directly contributed to the label definition.
- After documenting the Logistic Regression baseline, compare it with one shallow Decision Tree while keeping interpretability as the main priority.
- For future screening analyses, consider unequal error costs and report uncertainty around High Risk recall rather than relying on a single point estimate.

10. Conclusion

This project applies a transparent workflow to maternal health risk stratification using data quality review, exploratory statistics, standardized group comparisons, and multinomial Logistic Regression. In the 451 observation modeling sample, Blood Sugar showed the strongest separation for High Risk, followed by the blood pressure variables. The baseline model achieved 67.0% accuracy, but performance differed markedly across classes. Recall was 56.5% for High Risk and only 4.8% for Mid Risk.

The main conclusion is methodological. Overall accuracy is not sufficient for judging this model. Preprocessing assumptions, class specific recall, false negative errors, label ambiguity, and the limited set of predictors all affect how the results should be interpreted. The current Logistic Regression model is suitable as an educational baseline only and should not be used for diagnosis or clinical decision making.

References

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